

MA 102 (Mathematics II - Ordinary Differential Equations)

Department of Mathematics, IIT Guwahati

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Tutorial Sheet No. 2

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Lectures 5-6: Existence and uniqueness of solutions, applications of first-order ODEs.

1. Discuss the existence and uniqueness of the solution of the following initial value problems.

(i) $\frac{dy}{dx} = y^{1/3}, y(0) = 0.$

(ii) $\frac{dy}{dx} = y^{1/3}, y(0) = 1.$

(iii) $\frac{dy}{dx} = x^2 + y^2, y(0) = 0.$

(iv) $\frac{dy}{dx} = \sqrt{|1 - y^2|}, y(\pi/2) = 1.$

(v) $\ln\left(\frac{dy}{dx}\right) = y^2, y(x_0) = y_0, \text{ where } x_0, y_0 \in \mathbb{R}.$

2. Find the functions y_1, y_2 , and y_3 using the Method of Successive Approximations for the differential equation $\frac{dy}{dx} = y - x + 1$, with the initial condition $y(0) = 0$. Then find the exact solution to this initial value problem by taking the limit of the sequence of approximations.
3. Find p such that the $\frac{dy}{dx} = y^p, y(0) = 0$ admits a unique solution.
4. Show that the solution $y(x)$ of the initial value problem $\frac{dy}{dx} = 2 + \sin y, y(0) = 0$ cannot vanish for $x > 0$.

5. Consider the IVP:

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0.$$

Let $f(x + T, y) = f(x, y)$ and $y(x_0 + T) = y_0$, where $f(x, y)$ satisfies the assumptions of Picard's theorem. Then prove that the solution y is also periodic of period T , i.e. $y(x + T) = y(x)$ for all x .

6. For the initial value problem

$$\frac{dy}{dx} = x + e^x \sin(xy), \quad y(0) = y_0 = 0,$$

estimate the variation of the solution in the interval $[0, 1]$ if y_0 is perturbed by 0.01.

7. Let $y(x, \lambda)$ denote the unique solution of the initial value problem

$$(IVP)_\lambda : \quad \frac{dy}{dx} = \lambda + \cos y, \quad y(0) = y_0 = 0.$$

Then obtain an upper estimate for $|y(x, \lambda_1) - y(x, \lambda_2)|$ in the interval $[0, 1]$.

8. Show that the orthogonal trajectories of a given family of curves $f(x, y) = c$ are represented by the differential equation

$$\frac{dy}{dx} = \frac{\partial f}{\partial y} \bigg/ \frac{\partial f}{\partial x}.$$

9. Show that the family of parabolas $y^2 = 2cx + c^2$ where $c \in \mathbb{R}$ is an arbitrary constant is self-orthogonal. (A given family of curves is said to be self-orthogonal if its family of the orthogonal trajectories is the same as the given family.)
10. Find the value of n such that the curves $x^n + y^n = c_1$ are the orthogonal trajectories of the family $y = \frac{x}{1 - c_2 x}$, where c_1 and c_2 are arbitrary constants.
11. An object cools in air of constant temperature 20°C . Ten minutes after the object began to cool, its temperature was observed to be 75°C , and ten minutes later, its temperature was observed to be 50°C . What was its temperature when it began to cool?
12. Follow the current flow in the RL circuit:

$$L \frac{dI}{dt} + RI = E(t).$$

Find current $I(t)$ when the initial current is given as I_0 , and the emf imposed at time $t = 0$ is given by

- (a) $E = E_0 e^{-kt}$, $k > 0$,
(b) $E = E_0 \sin \omega t$.

The End